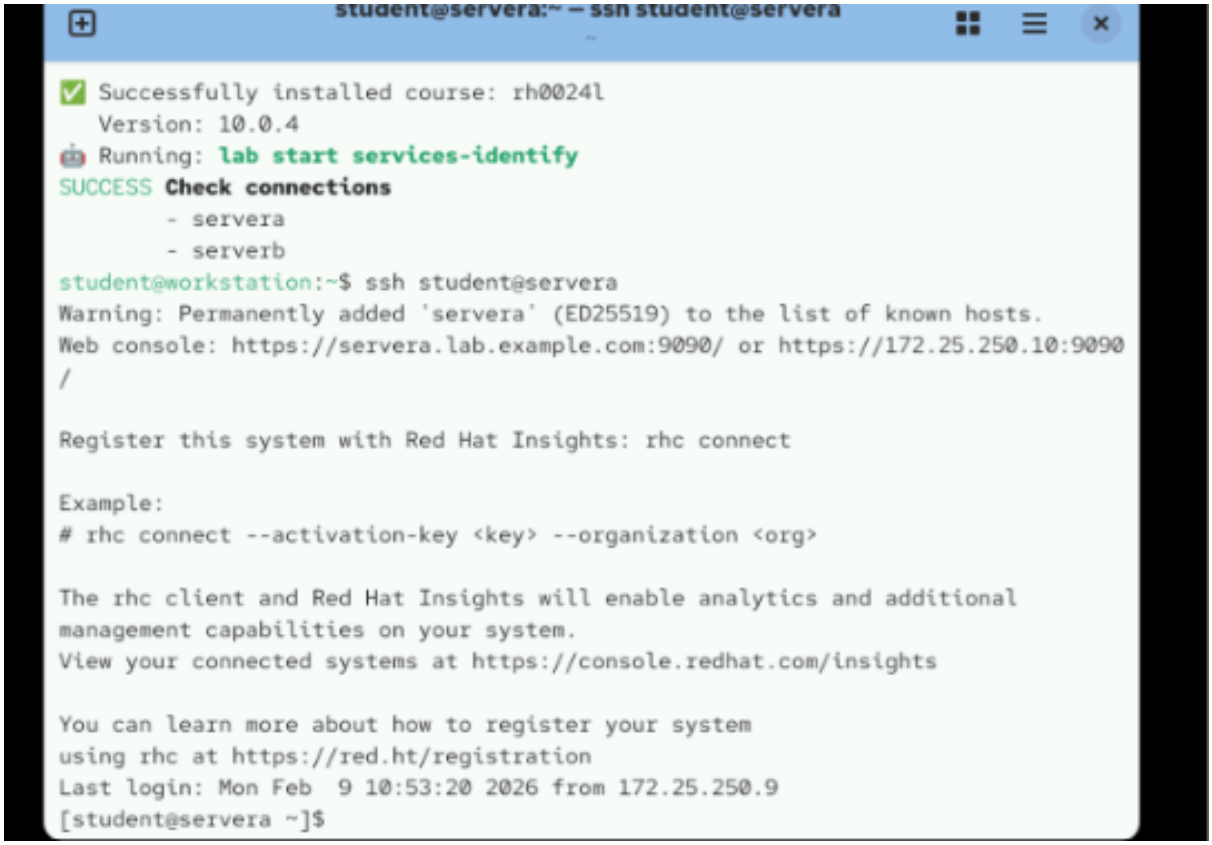
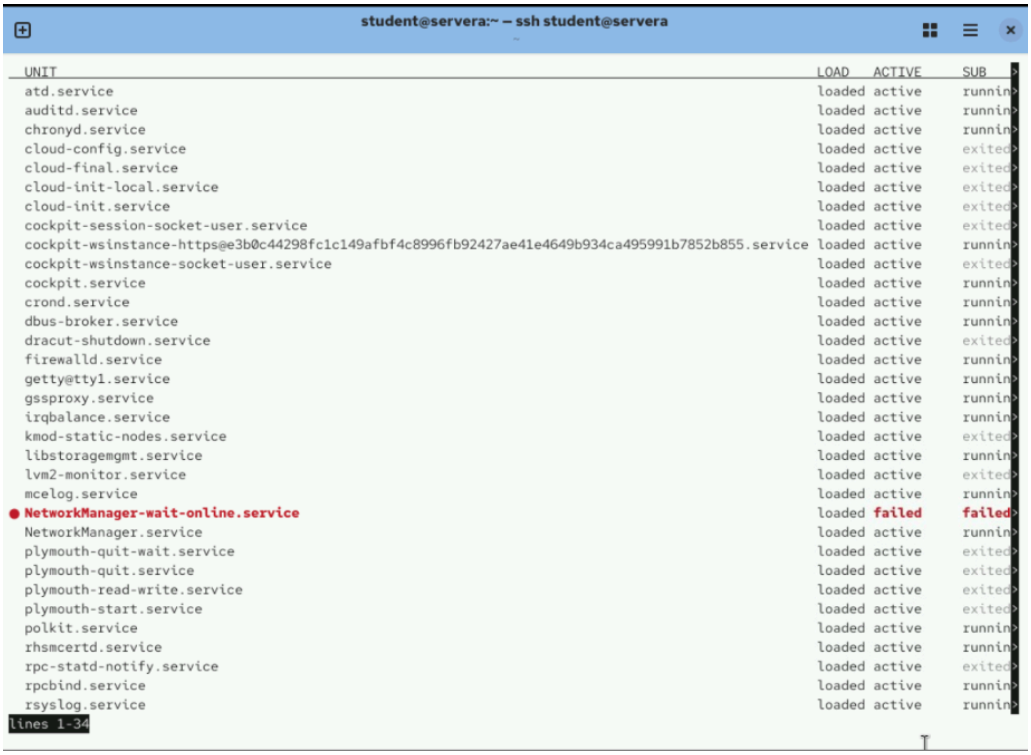
16.2-Guided Exercise: Identify System Processes that Start Automatically

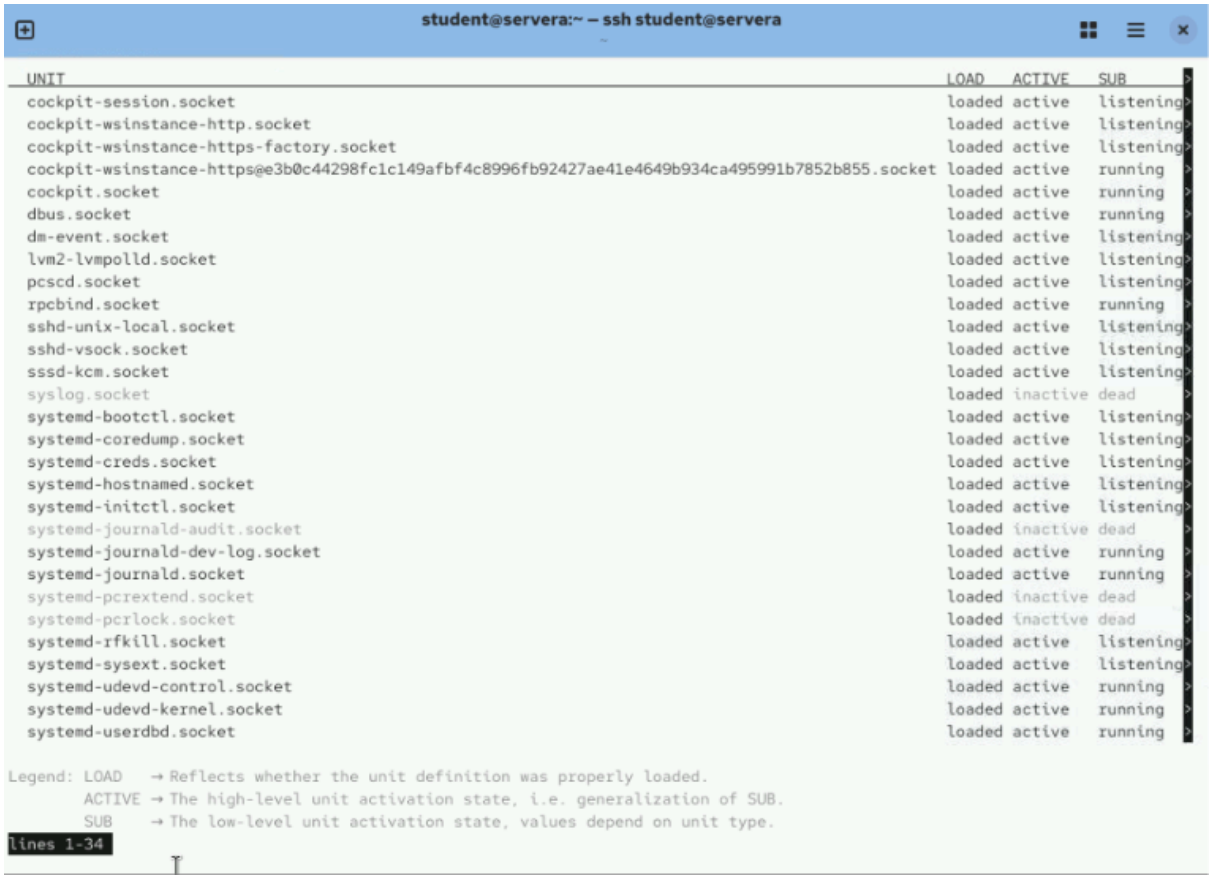
Ahmed Abdelhedi

1/




student@servera:~ — ssh student@servera
UNIT LOAD ACTIVE SUB
atd.service loaded active runnin
auditd.service loaded active runnin
chronyd.service loaded active runnin
cloud-config.service loaded active exited
cloud-final.service loaded active exited
cloud-init-local.service loaded active exited
cloud-init.service loaded active exited
cockpit-session-socket-user.service loaded active exited
cockpit-wsinstance-https@e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.service loaded active runnin
cockpit-wsinstance-socket-user.service loaded active exited
cockpit.service loaded active runnin
crond.service loaded active runnin
dbus-broker.service loaded active runnin
dracut-shutdown.service loaded active exited
firewalld.service loaded active runnin
getty@tty1.service loaded active runnin
gssproxy.service loaded active runnin
irqbalance.service loaded active runnin
kmod-static-nodes.service loaded active exited
libstoragemgmt.service loaded active runnin
lvm2-monitor.service loaded active exited
mcelog.service loaded active runnin
● NetworkManager-wait-online.service loaded failed failed
NetworkManager.service loaded active runnin
plymouth-quit-wait.service loaded active exited
plymouth-quit.service loaded active exited
plymouth-read-write.service loaded active exited
plymouth-start.service loaded active exited
polkit.service loaded active runnin
rhsmcertd.service loaded active runnin
rpc-statd-notify.service loaded active exited
rpcbind.service loaded active runnin
rsyslog.service loaded active runnin
lines 1-34

2/


student@servera:~ — ssh student@servera
UNIT LOAD ACTIVE SUB
cockpit-session.socket loaded active listening
cockpit-wsinstance-http.socket loaded active listening
cockpit-wsinstance-https-factory.socket loaded active listening
cockpit-wsinstance-https@e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.socket loaded active running
cockpit.socket loaded active running
dbus.socket loaded active running
dm-event.socket loaded active listening
lvm2-lvmpolld.socket loaded active listening
pcscd.socket loaded active listening
rpcbind.socket loaded active running
sshd-unix-local.socket loaded active listening
sshd-vsock.socket loaded active listening
sssd-kcm.socket loaded active listening
syslog.socket loaded inactive dead
systemd-bootctl.socket loaded active listening
systemd-coredump.socket loaded active listening
systemd-creds.socket loaded active listening
systemd-hostnamed.socket loaded active listening
systemd-initctl.socket loaded active listening
systemd-journald-audit.socket loaded inactive dead
systemd-journald-dev-log.socket loaded active running
systemd-journald.socket loaded active running
systemd-pcrextend.socket loaded inactive dead
systemd-pcrlock.socket loaded inactive dead
systemd-rfkill.socket loaded active listening
systemd-sysext.socket loaded active listening
systemd-udevd-control.socket loaded active running
systemd-udevd-kernel.socket loaded active running
systemd-userdbd.socket loaded active running
Legend: LOAD → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
SUB → The low-level unit activation state, values depend on unit type.
lines 1-34

3/

```
[student@servera ~]$ systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-09-28 15:48:43 UTC; 9min ago
 Invocation: e8598c798c174ee5861d58089ea844d9
       Docs: man:chronyd(8)
             man:chrony.conf(5)
    Process: 951 ExecStart=/usr/sbin/chronyd $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 974 (chronyd)
      Tasks: 1 (limit: 22366)
     Memory: 4.4M (peak: 5.3M)
        CPU: 57ms
     CGroup: /system.slice/chronyd.service
             └─974 /usr/sbin/chronyd -F 2

Sep 28 15:48:43 servera chronyd[974]: chronyd version 4.6.1 starting (+CMDMON +NTP +REFCLOCK +RTC +PRIVDROP +SCFILTER +SIGND>
Sep 28 15:48:43 servera chronyd[974]: commandkey directive is no longer supported
Sep 28 15:48:43 servera chronyd[974]: generatecommandkey directive is no longer supported
Sep 28 15:48:43 servera chronyd[974]: Could not open keyfile /etc/chrony.keys
Sep 28 15:48:43 servera chronyd[974]: Frequency -12.496 +/- 0.280 ppm read from /var/lib/chrony/drift
Sep 28 15:48:43 servera chronyd[974]: Loaded seccomp filter (level 2)
Sep 28 15:48:43 servera systemd[1]: Started chronyd.service - NTP client/server.
Sep 28 15:48:45 servera chronyd[974]: Source 172.25.254.254 offline
Sep 28 15:48:45 servera chronyd[974]: Source 172.25.254.254 online
Sep 28 15:48:49 servera chronyd[974]: Selected source 172.25.254.254
lines 1-24/24 (END)
```

```
[student@servera ~]$ ps -p 974
    PID TTY          TIME CMD
    974 ?        00:00:00 chronyd
[student@servera ~]$
```

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```
[student@servera ~]$ systemctl is-enabled sshd
enabled
[student@servera ~]$ systemctl is-active sshd
active
[student@servera ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-09-28 15:49:45 UTC; 11min ago
 Invocation: 3b63d6bd2acf41e48d6f45197b38a9b1
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1428 (sshd)
      Tasks: 1 (limit: 22366)
     Memory: 6.6M (peak: 7.8M)
        CPU: 84ms
     CGroup: /system.slice/sshd.service
             └─1428 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 28 15:49:45 servera systemd[1]: Starting sshd.service - OpenSSH server daemon...
Sep 28 15:49:45 servera (sshd)[1428]: sshd.service: Referenced but unset environment variable evaluates to an empty string: >
Sep 28 15:49:45 servera sshd[1428]: Server listening on 0.0.0.0 port 22.
Sep 28 15:49:45 servera sshd[1428]: Server listening on :: port 22.
Sep 28 15:49:45 servera systemd[1]: Started sshd.service - OpenSSH server daemon.
Sep 28 15:51:14 servera sshd-session[1535]: Accepted publickey for root from 172.25.250.9 port 42760 ssh2: RSA SHA256:M8ikhc>
Sep 28 15:51:14 servera sshd-session[1535]: pam_unix(sshd:session): session opened for user root(uid=0) by root(uid=0)
Sep 28 15:51:14 servera sshd-session[1535]: pam_unix(sshd:session): session closed for user root
Sep 28 15:52:06 servera sshd-session[1612]: Accepted publickey for student from 172.25.250.9 port 46526 ssh2: RSA SHA256:M8i>
Sep 28 15:52:06 servera sshd-session[1612]: pam_unix(sshd:session): session opened for user student(uid=1000) by student(uid>
lines 1-23/23 (END)
```

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```
[student@servera ~]$ systemctl list-unit-files --type=service
UNIT FILE                                          STATE          PRESET
arp-ethers.service                                 disabled       disabled
atd.service                                        enabled        enabled
audit-rules.service                                enabled        enabled
auditd.service                                     enabled        enabled
auth-rpcgss-module.service                         static         -
autovt@.service                                    alias          -
blk-availability.service                           disabled       disabled
bluetooth.service                                  enabled        enabled
capsule@.service                                   static         -
chrony-wait.service                                disabled       disabled
chronyd-restricted.service                         disabled       disabled
chronyd.service                                    enabled        enabled
cloud-config.service                               enabled        enabled
cloud-final.service                                enabled        enabled
cloud-init-hotplugd.service                        static         -
cloud-init-local.service                           enabled        enabled
cloud-init.service                                 enabled        enabled
cockpit-issue.service                              static         -
cockpit-session-socket-user.service                static         -
cockpit-session@.service                           static         -
cockpit-wsinstance-http.service                    static         -
cockpit-wsinstance-https-factory@.service          static         -
cockpit-wsinstance-https@.service                  static         -
cockpit-wsinstance-socket-user.service             static         -
cockpit.service                                    static         -
com.redhat.Yggdrasil1.Worker1.package_manager.service disabled    disabled
console-getty.service                              disabled       disabled
container-getty@.service                           static         -
crond.service                                      enabled        enabled
dbus-broker.service                                enabled        enabled
dbus-org.bluez.service                             alias          -
```

6/

```
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$ lab finish services-identify
🤖 Running: lab finish services-identify
student@workstation:~$
```